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PI Name:	Loucas, Bradford Ph.D.		
Project Title:	The Contribution of Delta Rays to the Chromosome Aberration Dose Response in Human Cells Irradiated with HZE Particles of Different Energy but the Same LET		
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Space Biology Special Category:	None		
PI Email:	avondeak@nasaprs.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	409-772-9745
Organization Name:	University of Texas Medical Branch, Galveston		
PI Address 1:	Radiation Oncology		
PI Address 2:	301 University Blvd, 0884		
PI Web Page:			
City:	Galveston	**State:**	TX
Zip Code:	77555-5302	**Congressional District:**	14
Comments:			
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Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
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COI Name (Institution):	Cornforth, Michael Ph.D. (University of Texas Medical Branch, Galveston)		
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	Highly energetic heavy ions (HZE), of the type found in galactic cosmic rays, deposit energy in the form of ionizations along the tracks they take as they travel through a cell. These ionizations are tightly spaced along these tracks and this spacing makes them highly damaging to cellular structures. Some of the particles' energy is also imparted to electrons (delta rays) which can travel great distances and produce damage, albeit less severe, in cells far removed from the original particle track. The ranges of these electrons are proportional to the energy of the original HZE particle. While these delta rays by themselves are not likely to produce much in the way of measurable damage, we have evidence that these electrons can add damage to that forming along independent HZE particle tracks that will eventually be expressed in the form of chromosome exchanges of greater complexity. Chromosome exchanges result when radiation breaks a chromosome by severing the DNA of which it is composed. The cell has means to repair the chromosome and can splice		

Task Description:	the DNA together. On some occasions, if one chromosome break is close to another break, the cell might make a mistake and join the broken ends to inappropriate partners. Two or more breaks can be involved in the process; the greater the number of breaks, the more complicated the resulting chromosome exchange will be. We plan to test the hypothesis that these exchange complexity increases will be related to the primary particle energy and resultant maximum delta ray range. Higher particle energies will result in longer delta ray maximum ranges allowing more primary particle tracks to be close enough to irradiate a cell with delta electrons for any specific fluence (particle tracks per unit area; equivalent to dose). We plan to irradiate human skin cells with low and high energy beams matching LET (linear energy transfer; energy deposited per unit track length) and fluences so that the only variable will be particle energy. A dose response curve for chromosome exchanges irradiated with low particle fluences will be constructed. At the same time we will also look for breaks that fail to rejoin. One experiment will be conducted at low dose rate which will prevent damage from delta rays from being added to chromosome exchanges forming along primary particle tracks by spreading damage formation out in time and allowing some breaks to be repaired or exchanges to be completely formed before other breaks occurring later can be added to them. We also plan to grow populations from single surviving cells allowing us to measure the transmissibility of chromosome exchanges a number of cell generations removed from the irradiation. Chromosome aberrations are tightly linked to cancer development. With this study we hope to gain a better understanding of the processes underlying chromosome exchange formation following irradiation with HZE particles similar to those astronauts might encounter in space. We plan to determine the shape of the exchange dose response curve for these particles at low doses and at low dose rate, processes involving the misrepair of radiation induced damage that could lead to carcinogenesis. At the same time we will look at terminal deletions and incomplete exchanges resulting from the cell's failure to repair damage. These lesions are likely to kill the cell and eliminate it from a population that might develop into cancer. Finally, we will look for exchanges found only in a fraction of the cells making up a clonal population indicating genomic instability, the process which drives carcinogenesis. This project will endeavor to gain understanding on how delta ray interactions with primary particle tracks might affect these endpoints that are associated with processes that lead to cancer development and enable NASA to produce more accurate models for cancer risk estimates from exposure to galactic cosmic rays.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The overarching goal of this proposal is to determine the effect of a single charged particle track passing through a cell nucleus without any influence from d-rays contributed by independent tracks passing through the medium at the same time. This is the condition likely to occur in space and understanding the effects produced under these conditions is important in order to assess the dangers associated with galactic cosmic rays (GCR). Regrettably, simulating these conditions in the Earth environment is virtually impossible as even at the smallest doses, effects of a single track will be impacted by d-rays originating from independent tracks. It is the goal of this study to measure these track interaction effects on the chromosome exchange breakpoint dose response in order to separate them from those produced by single primary tracks. A thorough understanding of the mechanisms that underlie track interaction should allow the construction of more precise biophysical models of the effects of HZE particles on living organisms thereby reducing the uncertainty in cancer risk estimates NASA seeks in order to assess the hazards of GCR. In addition, track interaction is likely to have effects on particle beam therapy giving this study further societal and economic impact.
Task Progress:	Our study involves the measurement of radiation induced chromosome aberrations produced by high energy heavy ions similar to those that are found in galactic cosmic rays. Chromosome aberrations are important radiation lesions that are associated with carcinogenesis. Chromosome aberrations are formed when radiation or some other treatment breaks chromosomes. Normally, the cell has the means to repair these breaks and joins the chromosomes back together. On occasion, if a number of breaks are close to one another, the cell can make a mistake and join the end one side of a break (break end) on one chromosome to one on another chromosome while the other break end from the first chromosome often is joined to the remaining break end on the second chromosome. This leads to an exchange of chromosomal segments that can be detected with chromosome analysis procedures. While many times, as in this example, exchanges involve only two breaks (termed simple exchanges) other times, three or more breaks may be involved in an exchange. For example, a break end from chromosome "A" can join to one from chromosome "B" while the other side of the break from "B" can join to one of the ends of a break on chromosome "C." The remaining break end from chromosome "C" can join to the open break end from chromosome "A" or it may join to a break end from on a fourth chromosome and so on. When three or more breaks are involved, the exchange is described as a "complex exchange." Since our last report, we have been busy karyotyping (generating chromosome profiles) of the cells irradiated so far. These include fibroblasts (skin cells) and lymphocytes (blood cells) irradiated with iron or silicon ions. We were also able to irradiate cells with calcium ions on April 14 of this year. This experiment worked well for the fibroblasts but we had difficulty collecting sufficient numbers of analyzable lymphocytes. In addition, were able to freeze 19 cell populations, each derived from a single cell that survived the irradiation. These populations will eventually be thawed and examined for chromosomal changes that might have occurred at some time after the irradiation which would be indicative of chromosomal instability, a potential hallmark for carcinogenesis. In addition we have begun discussions with collaborators concerning special statistical methods that might provide better procedures for analyzing our data.
Bibliography Type:	Description: (Last Updated: 10/29/2023)
Abstracts for Journals and Proceedings	Loucas BD, Cornforth MN. "The production of chromosomal exchanges by ions of different energies but having the same LET." Poster presentation at the 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. , Feb-2016
Abstracts for Journals and Proceedings	Loucas BD, Cornforth MN. "The production of chromosomal exchanges by ions of different energies but having the same LET." Poster presentation at 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. , Oct-2016

Abstracts for Journals and Proceedings	Shuryak I, Loucas BD, Cornforth MN. "Seeking beta: Experimental considerations and the theoretical implications regarding the detection in dose response relationships for true simple exchanges." Poster presentation at 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. 62nd Annual Meeting of the Radiation Research Society, Big Island, Hawaii, October 16-19, 2016. , Oct-2016
Articles in Peer-reviewed Journals	Loucas BD, Shuryak I, Cornforth MN. "Three-color chromosome painting as seen through the eyes of mFISH: Another look at radiation-induced exchanges and their conversion to whole-genome equivalency." Frontiers in Oncology. 2016 Mar 15;6:52. eCollection 2016. http://dx.doi.org/10.3389/fonc.2016.00052 ; PubMed PMID: 27014627; PubMed Central PMCID: PMC4791380 , Mar-2016
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